Experiment 18

Optical Spectrum and OSNR Analysis using OptiSystem

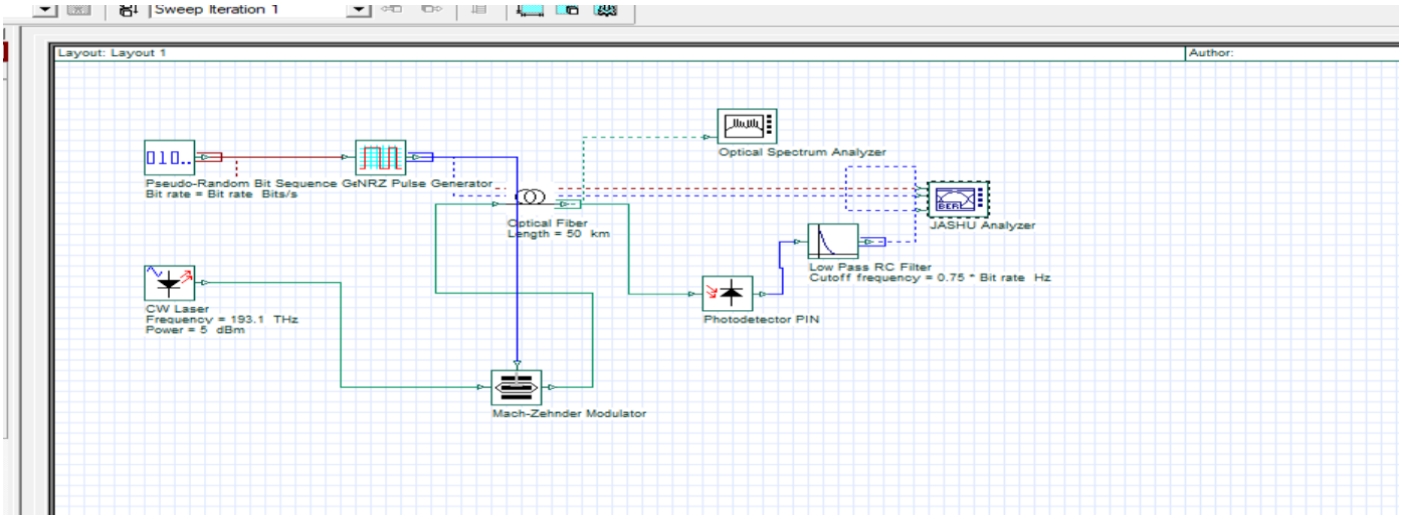
OBJECTIVES:

1. Design an optical communication system using OptiSystem.
2. Study how source power affects optical spectral intensity.
3. Analyze how fiber length affects attenuation, dispersion, and nonlinear effects.
4. Study the combined effect of source power and fiber length.

SOFTWARE USED:

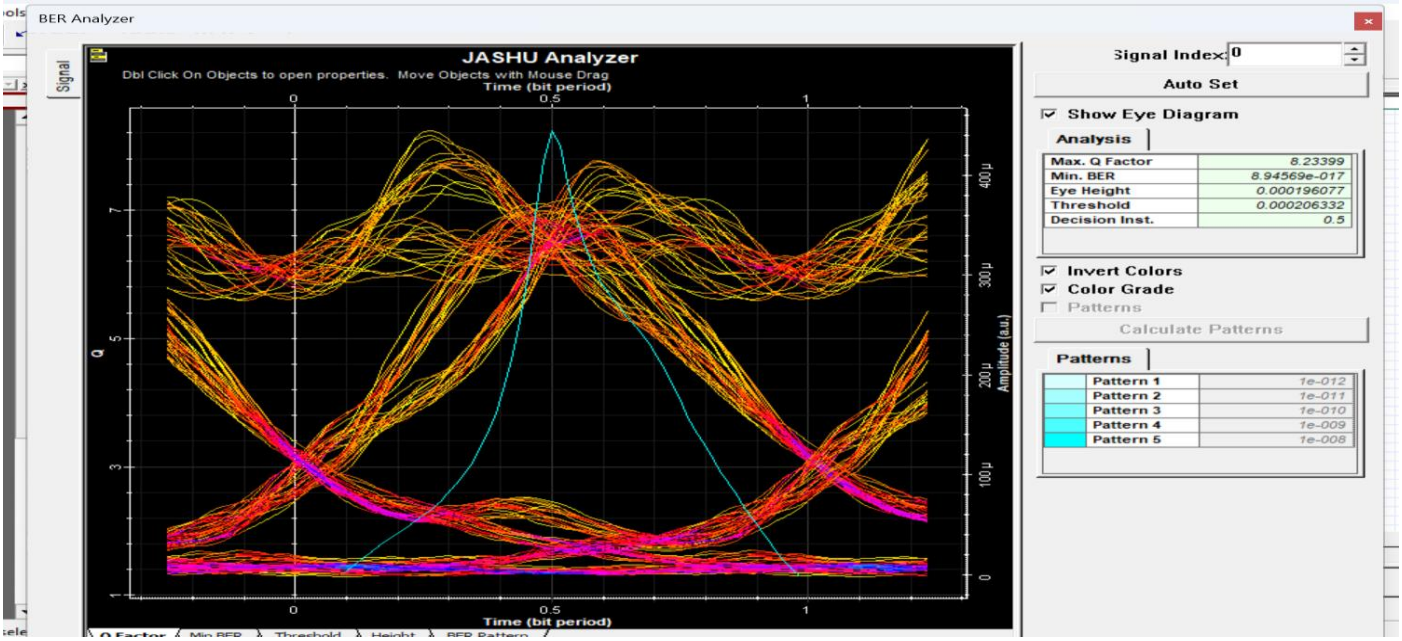
- Optisystem Software

BLOCK DIAGRAM

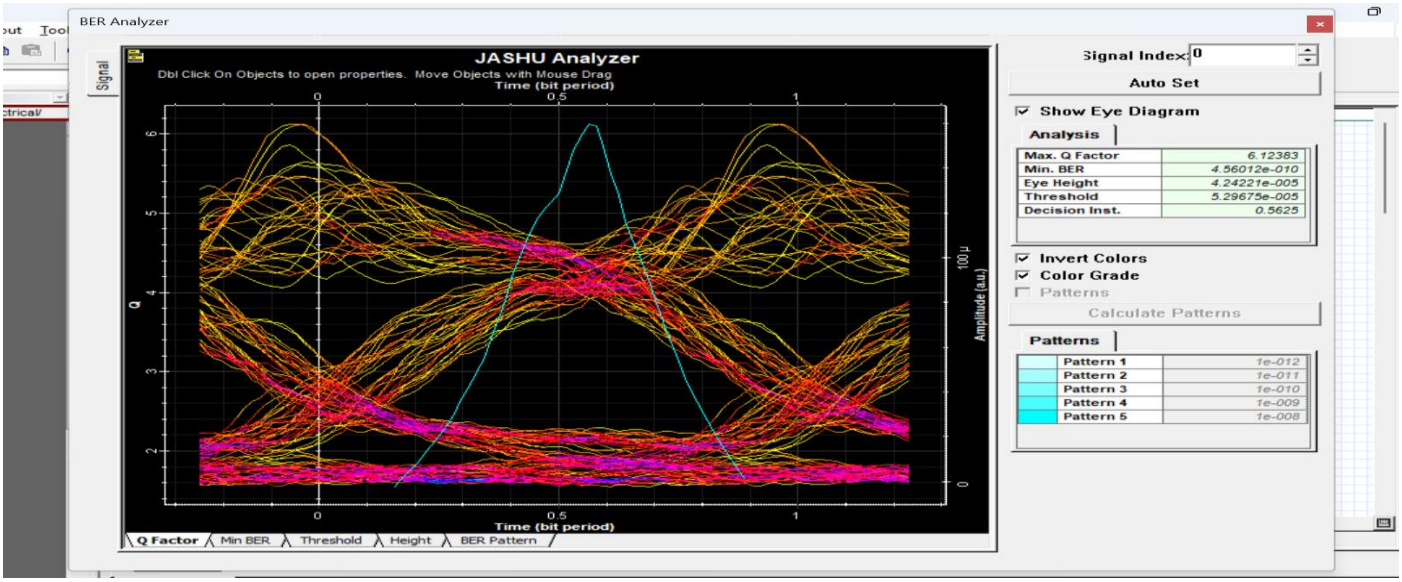


SAMPLE OUTPUT

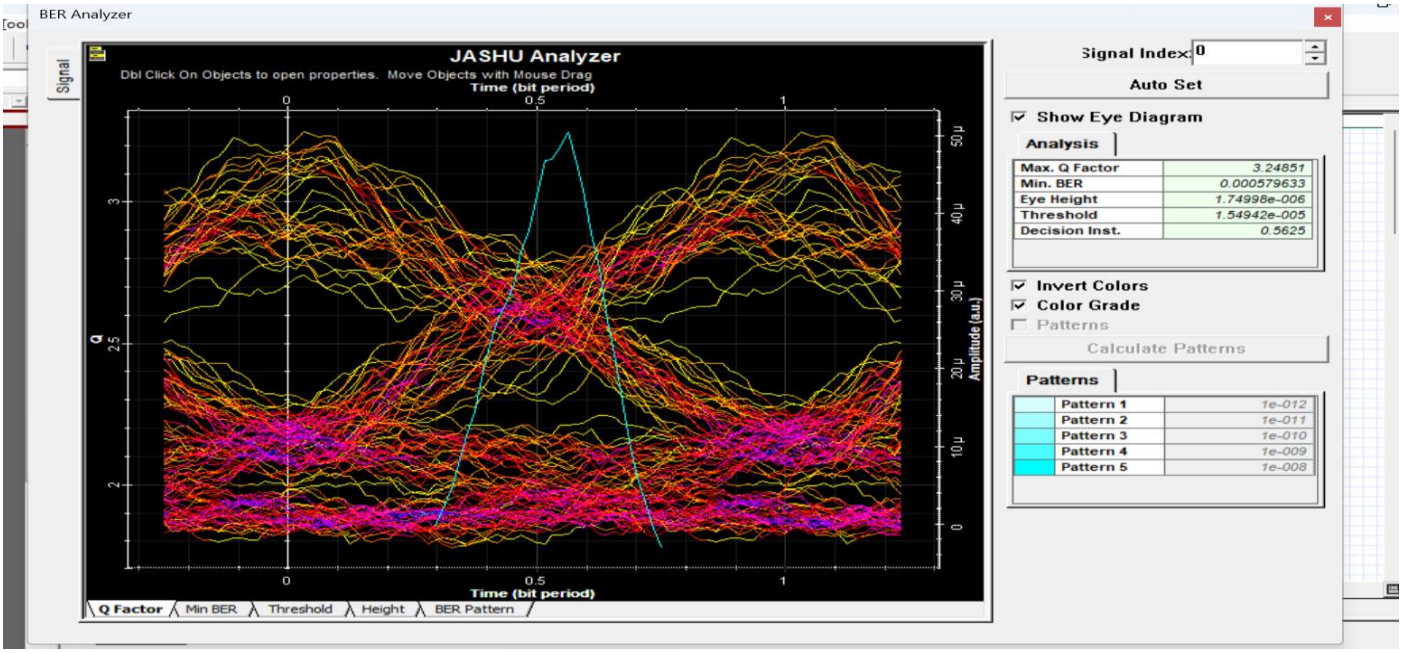
Power=5 dbm , Length= 50km



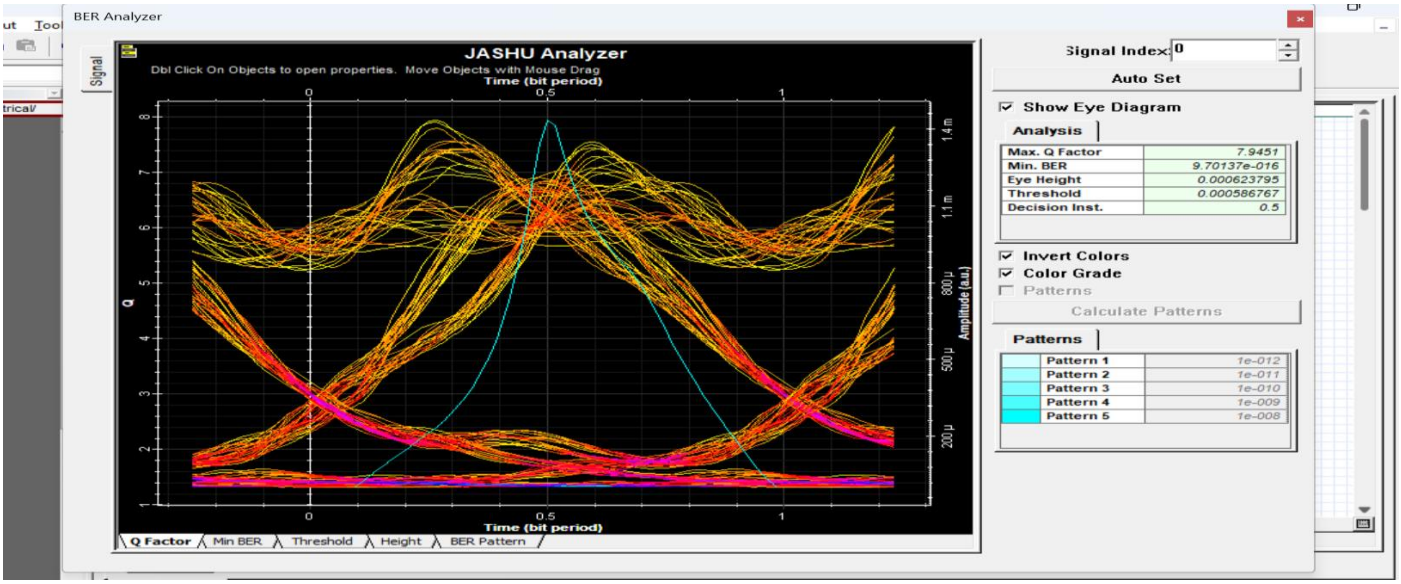
Power=5 dbm , Length= 75km



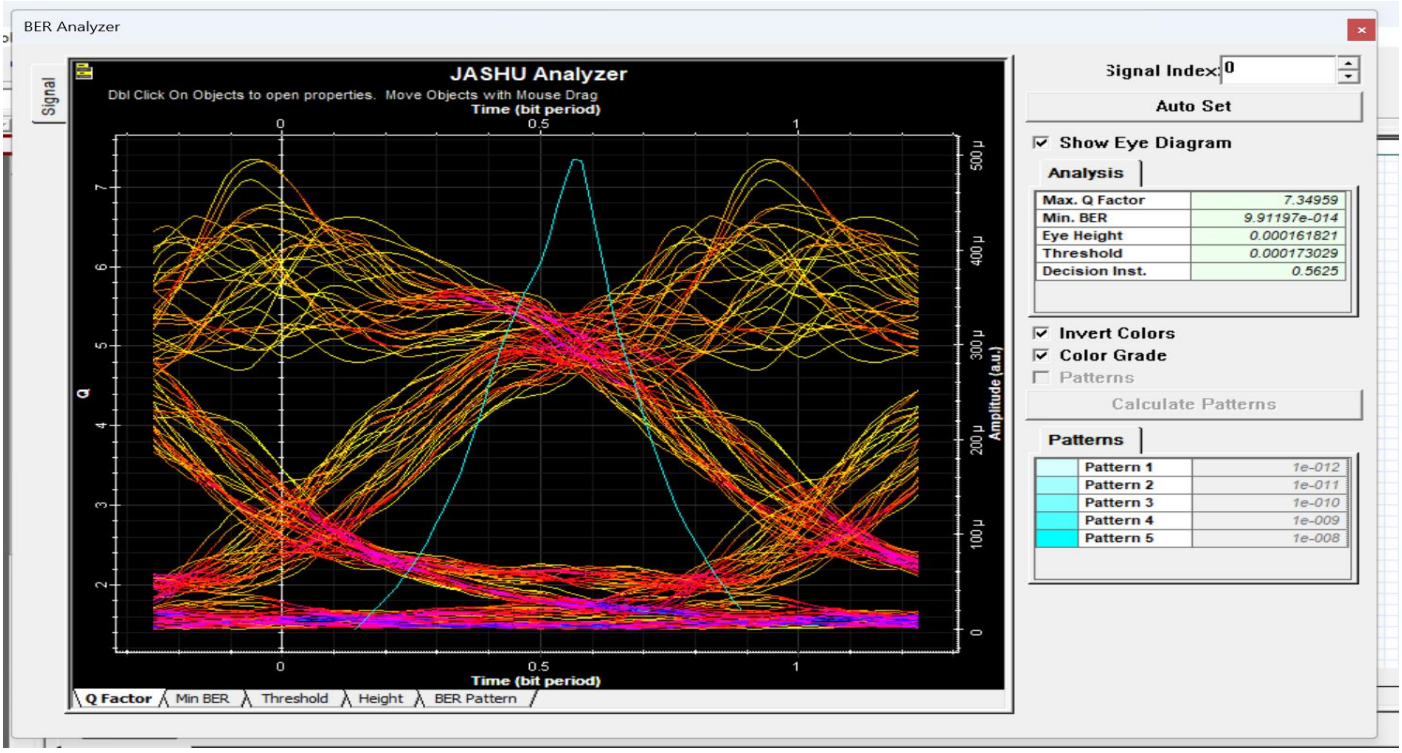
Power=5 dbm , Length= 100km



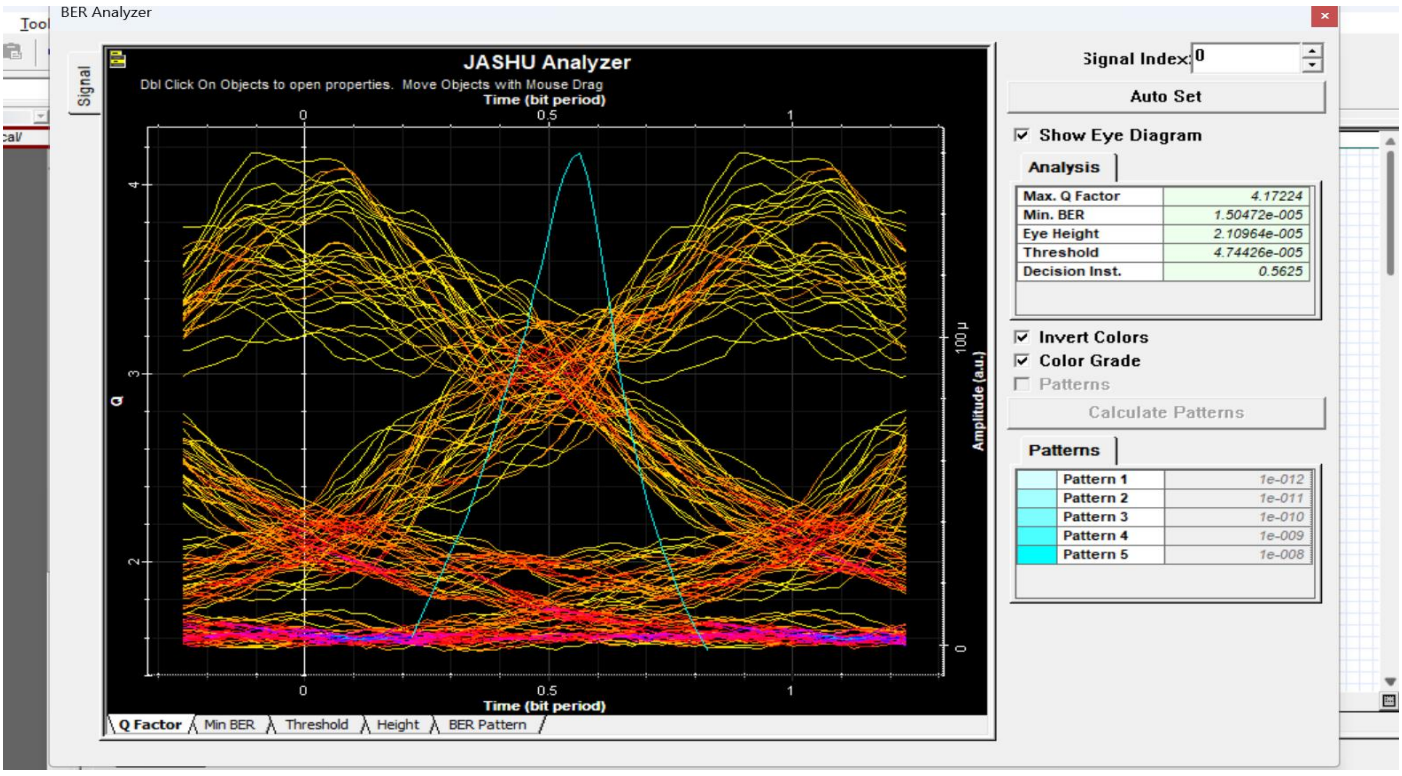
Power=10dbm , Length= 50km



Power=10dbm , Length= 75km



Power=10dbm , Length= 100km



Reflection: I learned how to observe optical signal power at different wavelengths. I understood the importance of OSNR in evaluating optical signal quality.

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

S.No.	Assessment Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Weightage
1	System Design & Simulation Setup	All OptiSystem components are correctly selected, connected and configured without errors.	All components are present with minor connection or layout errors.	Component selection is partially complete; minor setup errors affect signal flow.	Incorrect component selection or improper setup leads to simulation failure.	20%
2	Parameter Configuration & Execution	All input parameters and test cases (1, 2 and 3) are accurately configured and executed.	Parameters are mostly accurate with minor discrepancies.	Basic parameters are configured, but some test case settings are missing.	Incorrect parameter settings or failure to execute required test cases.	20%
3	Measurement & Data Collection	Accurate data collection for Min BER, Max Q-factor, spectral power and eye diagrams across all test cases.	Data is collected with minor omissions in BER, Q-factor or spectral readings.	Partial data collection with incomplete tables or missing eye diagram readings.	Inaccurate or missing measurements and incomplete data.	20%
4	Data Analysis & Spectral Interpretation	Comprehensive analysis relating source power and fiber length to BER, Q-factor, attenuation and dispersion.	Clear analysis identifying basic trends in BER and Q-factor.	Basic analysis without explaining physical effects such as dispersion or non-linearity.	Incorrect or missing analysis of simulation results and graphs.	20%
5	Report & Conclusion	Clear graphs, complete tables and meaningful conclusions identifying optimal operating parameters.	Relevant graphs and a proper concluding summary are provided.	Incomplete report with missing graphs, diagrams or weak conclusions.	Disorganized report without required diagrams and valid conclusions.	20%
	Total					100%

Result: The optical spectrum and OSNR of the communication link were analyzed successfully using OptiSystem.